FIFTH SEMESTER (CUFYUGP) DEGREE EXAMINATION: MODEL PAPER 5

MATHEMATICS

REAL ANALYSIS II

Selected Problems on Riemann Integration, Sequences of Functions & Interchange of Limits

Maximum Time: 2 Hours		**Maximum Marks: 70**
	Section A	
	All Questions can be answered. Each Question carries 3 marks (Ceiling: 24 Marks)	
1	Let f be continuous on $[a,b]$ such that $f(x) \geq 0$ for all $x \in [a,b]$. Prove that if $\int_a^b f = 0$, then $f(x) = 0$ for all $x \in [a,b]$.	**3**
2	Let $f \in \mathcal{R}[a,b]$ and let $c \in \mathbb{R}$. Define g on $[a+c, b+c]$ by $g(y) = f(y-c)$. Prove that $g \in \mathcal{R}[a+c, b+c]$ and that $\int_{a+c}^{b+c} g = \int_a^b f$.	**3**
3	Define pointwise convergence of a sequence of functions (f_n) on a subset $A \subseteq \mathbb{R}$. Find the pointwise limit of $f_n(x) = \dfrac{x}{x+n}$ for $x \geq 0$.	**3**
4	Evaluate the pointwise limit of the sequence $f_n(x) = \dfrac{nx}{1+n^2x^2}$ for all $x \in \mathbb{R}$.	**3**
5	Define the uniform norm of a bounded function on a set $A \subseteq \mathbb{R}$. Calculate the uniform norm $\|f_n - f\|_A$ for $f_n(x) = \dfrac{x}{n}$ on $A = [0,1]$ where $f(x) = 0$.	**3**
6	State the Cauchy Criterion for the uniform convergence of a sequence of bounded functions on a set $A \subseteq \mathbb{R}$.	**3**
7	Give an example of a sequence of continuous functions on $[0,1]$ whose pointwise limit is discontinuous. State where the uniform convergence fails.	**3**
8	Show by a concrete example that the limit of an integral is not necessarily equal to the integral of the limit, i.e., $\lim_{n\to\infty} \int_a^b f_n(x)\,dx \neq \int_a^b \lim_{n\to\infty} f_n(x)\,dx$.	**3**
9	State the theorem on the interchange of limit and continuity for a sequence of functions. Is uniform convergence a necessary condition for the continuity of the limit function? Justify.	**3**
10	Let (f_n) be a sequence of continuous functions on an interval I converging uniformly on I to f. If $(x_n) \subseteq I$ converges to $x_0 \in I$, show that $\lim_{n\to\infty} f_n(x_n) = f(x_0)$.	**3**
	Section B	
	All Questions can be answered. Each Question carries 6 marks (Ceiling: 36 Marks)	
11	Let $f : [0,1] \to \mathbb{R}$ be continuous and suppose that $\int_0^x f = \int_x^1 f$ for all $x \in [0,1]$. Prove that $f(x) = 0$ for all $x \in [0,1]$.	**6**
12	Suppose f is bounded by M on $[a,b]$ and its restriction to $[c,b]$ is Riemann integrable for every $c \in (a,b)$. Using the Squeeze Theorem, prove that $f \in \mathcal{R}[a,b]$ and that $\lim_{c\to a^+} \int_c^b f = \int_a^b f$.	**6**
13	Let $f_n(x) = \dfrac{x^n}{1+x^n}$ for $x \geq 0$. Find the pointwise limit function $f(x)$ on $[0,\infty)$. Examine whether the convergence is uniform on $[0,1]$ and on $[2,\infty)$.	**6**
14	Prove that a sequence (f_n) of bounded functions on $A \subseteq \mathbb{R}$ converges uniformly on A to a function f if and only if $\|f_n - f\|_A \to 0$ as $n \to \infty$.	**6**
15	Let $f_n(x) = x + \dfrac{1}{n}$ and $f(x) = x$ for $x \in \mathbb{R}$. Show that (f_n) converges uniformly to f on $\mathbb{R}$, but the sequence of products (f_n^2) does not converge uniformly to f^2 on $\mathbb{R}$. What does this demonstrate about the product of uniformly convergent sequences?	**6**
16	Let $f_n : [0,1] \to \mathbb{R}$ be defined for $n \geq 2$ by: $$f_n(x) = \begin{cases} n^2 x, & 0 \leq x \leq 1/n \\ -n^2(x - 2/n), & 1/n \leq x \leq 2/n \\ 0, & 2/n \leq x \leq 1 \end{cases}$$ Find the pointwise limit $f(x) = \lim_{n\to\infty} f_n(x)$ on $[0,1]$, and evaluate both $\lim_{n\to\infty} \int_0^1 f_n(x)\,dx$ and $\int_0^1 f(x)\,dx$. Explain why term-by-term integration fails in this case.	**6**
17	Let (f_n) be a sequence of continuous functions on a set $A \subseteq \mathbb{R}$ and suppose that (f_n) converges uniformly on A to a function f. Prove that f is continuous on A.	**6**
18	State Dini's Theorem for uniform convergence of sequences of functions. Provide counterexamples showing that the theorem may fail if:	**6**

	(a) The sequence is not monotone, or (b) The underlying interval is not closed and bounded.	
Section C		
Answer any ONE. Each Question carries 10 marks (1x10=10 Marks)		
19	(a) State and prove the Squeeze Theorem for Riemann integrable functions on $[a, b]$. (b) Let f be continuous on $[a, b]$ and let g be continuous on $[a, b]$ with $g(x) > 0$ for all $x \in [a, b]$. Prove that there exists a point $c \in [a, b]$ such that $$\int_a^b f(x)g(x)\,dx = f(c)\int_a^b g(x)\,dx$$ Show by an example that the condition $g(x) > 0$ cannot be dropped.	**10**
20	(a) State and prove the Cauchy Criterion for the uniform convergence of a sequence of bounded functions on a set $A \subseteq \mathbb{R}$. (b) Consider the sequence $f_n(x) = \frac{nx}{1+n^2x^2}$ on $[0, \infty)$. Prove that (f_n) converges uniformly to 0 on $[a, \infty)$ for any $a > 0$, but does not converge uniformly on $[0, \infty)$.	**10**